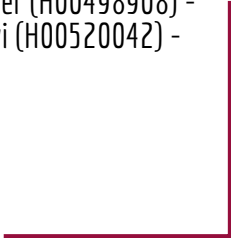




Youtube Video Popularity Prediction

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Research Question & Objective

Question

Can we predict the number of views a youtube video will get using the thumbnail, title/metadata/channel statistics?

Objective

Develop predictive models to estimate video popularity and identify which feature (thumbnail, metadata) most influence youtube video views.

→ Hypothesis

Videos with more visually appealing thumbnails and optimized metadata will achieve higher predicted view counts

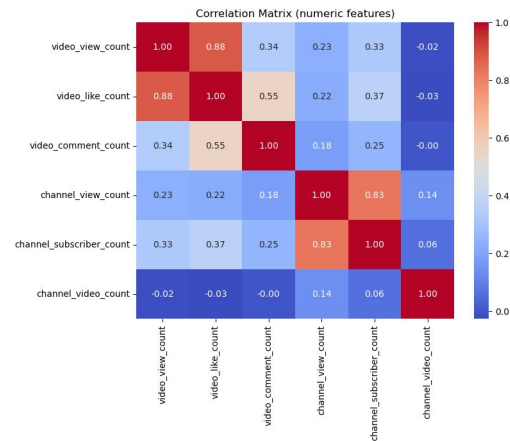
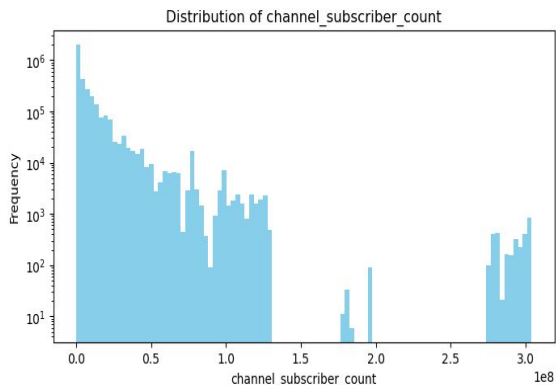
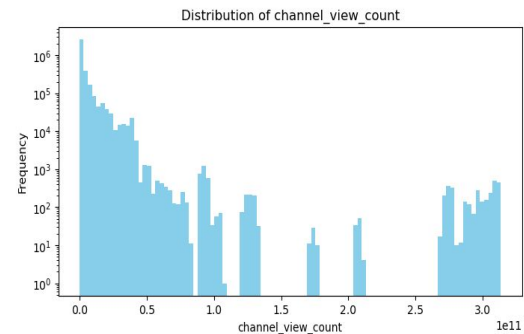
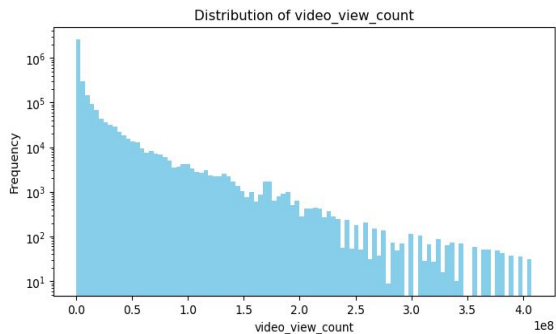
Datasets

- **Youtube API v3 Dataset Extraction (Image + Tabular)**
 - Fetching youtube thumbnails and metadata using the Youtube V3 API
 - 10,000 queries per day (estimated 5,000 videos per day)
 - 14 video attributes, and 9 channel attributes (which will be filtered)

API Call	Cost per call	Max results
channels.list	1 unit	50 channels per call
search.list	100 units	50 videos per call
videos.list	1 unit	50 videos per call

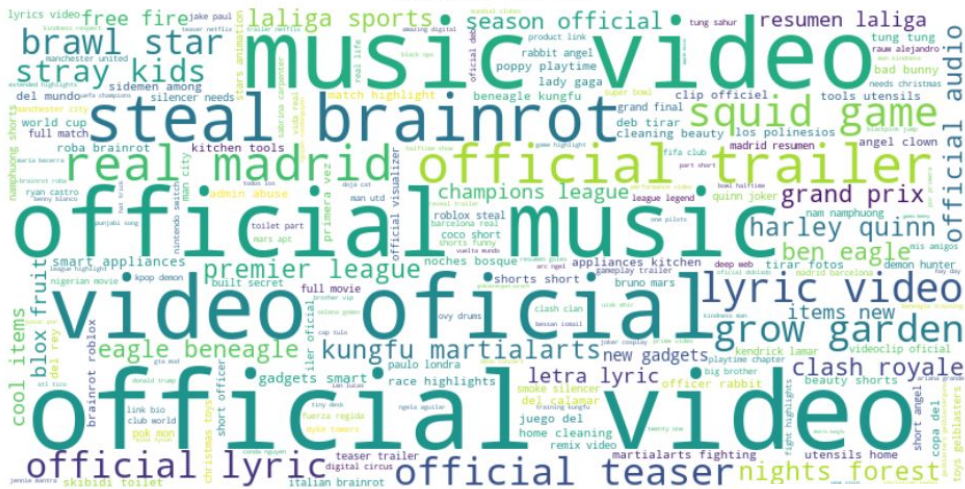
Attributes														
Video	Vid Id	Title	Description	Publish Date	Duration	View Count	Like Count	Comment Count	Favorite Count	Thumbnail Default URL	Thumbnail_Medium URL	Thumbnail High URL	Chan ID	Category ID
Channel	ID	Title	Description	Creation Date	Sub count	Video Count	View Count	Thumbnail URL	Thumbnail URL2					

Data Visualization*



* These visualizations are from the initial data pull, which will be further fine tuned to fit our model requirements

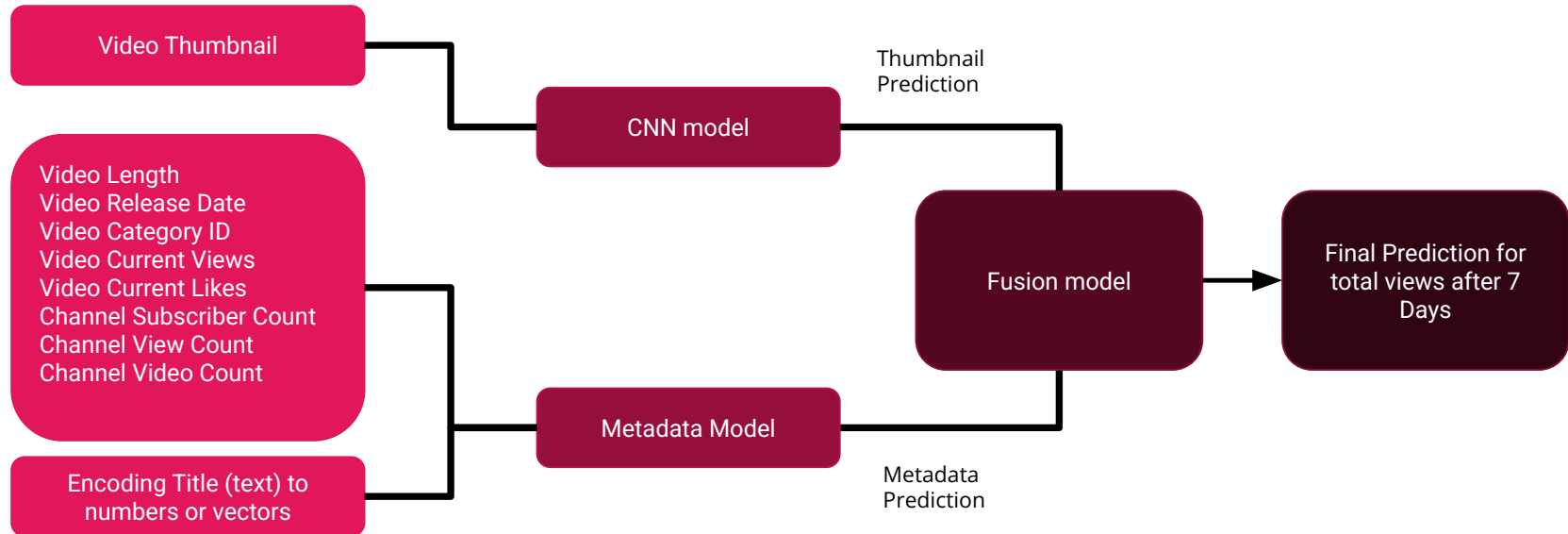
WordCloud of Video Titles



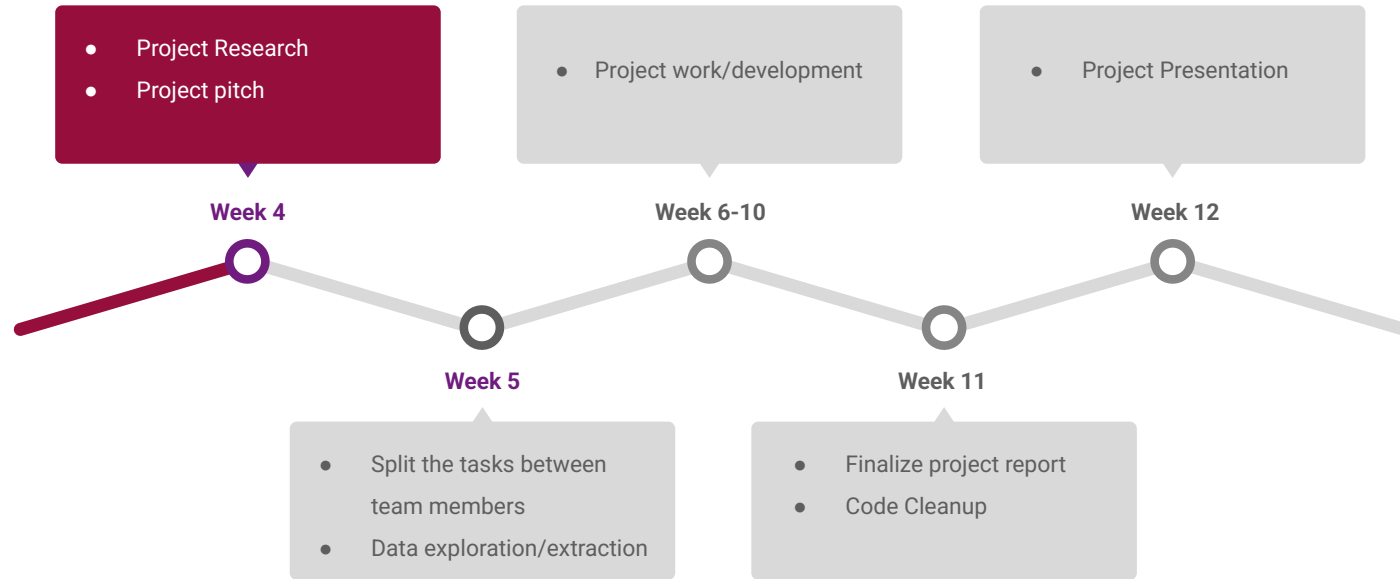
WordCloud of Video Descriptions



Methodology



Project Plan & Timeline



Expected Outcome

- Insight into which factor (Thumbnail, description, title.. etc) drives popularity the most.
- Visualisations of results
- A framework that would predict the youtube video views.



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Thank You